Mathematical and Computational Methods

Unit 9: Integration I: the definite integral and substitution

Units 7 and 8 built the derivative and put it to work. This unit starts the return journey. *Integration* answers a different question—not "how fast?" but "how much in total?"—and we meet it in two guises that turn out to be one. The *definite integral* appears first as the area under a curve, captured as a limit of sums; the *indefinite integral* appears as the antiderivative, the reverse of differentiation. The *Fundamental Theorem of Calculus* ties the two together and, remarkably, makes areas computable by antidifferentiation. We finish with the first and most important technique—*substitution*, which runs the chain rule of Unit 7 backwards.

Where this fits. The raw materials are from Units 7–8: every standard integral is a standard derivative read in reverse, substitution undoes the chain rule, and the continuity of Unit 7 is exactly the hypothesis the Fundamental Theorem needs. The area interpretation continues the curve-sketching of Unit 8—the definite integral measures the regions you learned to draw there. Everything here is the foundation of Unit 10, which adds the heavier techniques (by parts, partial fractions, trigonometric methods) and the applications (areas, volumes and expectations). Two further uses of integration—Taylor series and differential equations—belong to the dedicated Calculus module.

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Summary **13**

Learning objectives

By the end of this unit you will be able to:

- Read the definite integral as a Riemann-sum limit and as signed area.
- State and apply both parts of the Fundamental Theorem of Calculus.
- Recall and use the table of standard integrals.
- Integrate by substitution, including changing limits and the $f'/f \to \ln$ pattern.
- Know that some integrals, such as $\int \mathrm{e}^{-x^2}\,\mathrm{d}x$, have no elementary antiderivative.

1 The integral

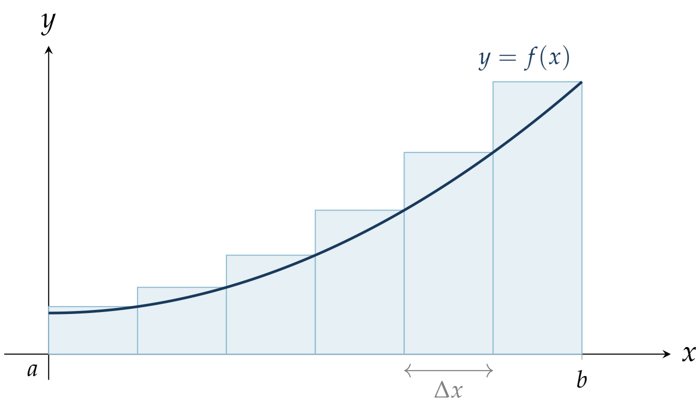
1.1 The area under a curve

We can find the area of a rectangle or a triangle from a formula, but what is the area of the region under a curved graph $y = f(x)$, above the x-axis, between $x = a$ and $x = b$? There is no elementary geometry formula for a curved region—so we approximate, and then improve the approximation without limit.

Cut the interval $[a, b]$ into n thin strips, each of width $\Delta x = \dfrac{b-a}{n}$. Over one strip the curve barely changes height, so the strip is almost a rectangle: width Δx, height $f(x_k)$ at any chosen point x_k of the k-th strip. (For a continuous f the limit below exists and is the same whatever points x_k are chosen.) Its area is about $f(x_k)\,\Delta x$, and the whole region is approximately the sum of these rectangles,

$$\text{area} \;\approx\; \sum_{k=1}^{n} f(x_k)\,\Delta x,$$

a *Riemann sum*. The picture below shows six such rectangles; already they trace the region, and with more, thinner strips the staircase hugs the curve ever more closely.



Example 1.1 (*Estimating the area by rectangles*). Take the curve $f(x) = 0.4 + 0.25x^2$ on $[0, 3]$ drawn above, cut into $n = 6$ strips of width $\Delta x = 0.5$. Using the height at the *right* end of each strip gives

$$\sum_{k=1}^{6} f(x_k)\,\Delta x = 0.5\big(0.4625 + 0.65 + 0.9625 + 1.4 + 1.9625 + 2.65\big) \approx 4.04,$$

an over-estimate, since on a rising curve the right end is the tallest point of each strip. Taking instead the *left*-hand heights gives ≈ 2.92, an under-estimate. The true area is therefore trapped between 2.92 and 4.04; halving the strip width tightens the bracket, and in the limit both estimates squeeze onto the exact value 3.45, which the Fundamental Theorem will deliver in one line in Section 2.

1.2 The definite integral

To get the *exact* area we let the strips become infinitely thin, $\Delta x \to 0$ (so $n \to \infty$). If the sums settle on a single number, that number is the definite integral.

Definition 1.1 (The definite integral). For a function f on $[a, b]$, the *definite integral* of f from a to b is the limit of its Riemann sums,

$$\int_a^b f(x)\,\mathrm{d}x = \lim_{\Delta x \to 0} \sum_{k=1}^{n} f(x_k)\,\Delta x,$$

when this limit exists (it always does if f is continuous on $[a, b]$). The numbers a and b are the *limits of integration*, f is the *integrand*, and the elongated S, $\int$, is Leibniz's sum sign.

The notation records the construction exactly: $\int$ is the limiting "sum," $f(x)$ is the height of a strip and $\mathrm{d}x$ its vanishing width, so $f(x)\,\mathrm{d}x$ is the area of one sliver and $\int_a^b$ adds the slivers from a to b.

Three properties can be read straight off the sums, because areas add. A strip of zero width has no area, so $\int_a^a f(x)\,\mathrm{d}x = 0$. Splitting $[a, b]$ at a point c splits the sum, so $\int_a^b f(x)\,\mathrm{d}x = \int_a^c f(x)\,\mathrm{d}x + \int_c^b f(x)\,\mathrm{d}x$. And we *define* $\int_b^a f(x)\,\mathrm{d}x = -\int_a^b f(x)\,\mathrm{d}x$, which keeps the splitting rule true whatever the order of a, b and c. We use these freely from now on; §2.3 collects them with linearity and checks them again.

Note (A definite integral is a number; the variable is a dummy). Once the limits are in place the result is a *number*, not a function of x. The integration variable is therefore a *dummy*: it may be renamed freely without changing the value,

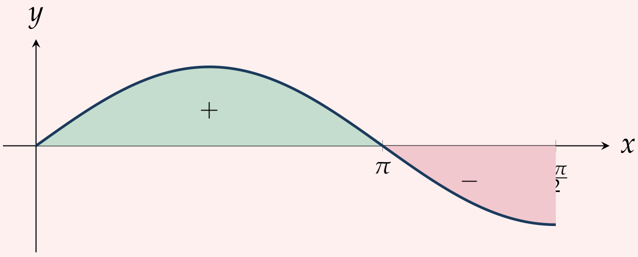
$$\int_a^b f(x)\,\mathrm{d}x = \int_a^b f(t)\,\mathrm{d}t = \int_a^b f(\xi)\,\mathrm{d}\xi.$$

A definite integral depends only on the integrand and the two limits.

Note (Signed area). The strip area $f(x)\,\Delta x$ is negative wherever $f(x) < 0$, so the definite integral counts area *below* the axis as negative and area *above* as positive. It returns the *net signed area*, not the total painted area. For the curve below,

$$\int_0^{3\pi/2} \sin x\,\mathrm{d}x = \underbrace{(+2)}_{[0,\pi]} + \underbrace{(-1)}_{[\pi,3\pi/2]} = 1.$$

(The values 2 and -1 are evaluated in §2 once we have the Fundamental Theorem; here only their signs matter.)



1.3 Computing an integral from the definition

Before the Fundamental Theorem rescues us, it is worth carrying the definition through *once* on a genuinely curved example, both to see that the limit really exists and to have something to check the shortcut against. Take the simplest curve that rectangle-and-triangle formulas cannot handle, $f(x) = x^2$ on $[0, 1]$.

Use the regular partition into n equal strips: the division points are $x_k = k/n$ for $k = 0, 1, \ldots, n$, each strip has width $\Delta x = 1/n$, and we read the height at the *right* endpoint $x_k = k/n$ of the k-th strip. The Riemann sum is then

$$S_n = \sum_{k=1}^{n} f\left(\frac{k}{n}\right) \Delta x = \sum_{k=1}^{n} \left(\frac{k}{n}\right)^2 \frac{1}{n} = \frac{1}{n^3} \sum_{k=1}^{n} k^2.$$

The sum that remains is a standard one—the next case after the $\sum_{k=1}^{n} k = \frac{1}{2}n(n+1)$ of Unit 1, provable by induction (Unit 3); we quote it:

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}.$$

Substituting and simplifying,

$$S_n = \frac{1}{n^3} \cdot \frac{n(n+1)(2n+1)}{6} = \frac{(n+1)(2n+1)}{6n^2} = \frac{1}{6}\left(1 + \frac{1}{n}\right)\left(2 + \frac{1}{n}\right).$$

This is an honest formula for the staircase area with n strips—for $n = 10$ it already gives $S_{10} = \frac{11 \cdot 21}{600} = 0.385$. Now let the strips become infinitely thin, $n \to \infty$: the two $1/n$ terms vanish and

$$\int_0^1 x^2 \, \mathrm{d}x = \lim_{n \to \infty} S_n = \frac{1}{6}(1)(2) = \frac{1}{3}.$$

The limit exists and equals exactly $\frac{1}{3}$. The same calculation on $[0, b]$—now with $\Delta x = b/n$ and $x_k = kb/n$—carries a factor b^3 through and yields $\int_0^b x^2 \, \mathrm{d}x = b^3/3$.

This is the value the Fundamental Theorem will produce in Section 2 in a single line, as $\left[x^3/3\right]_0^1 = \frac{1}{3}$. The agreement is the whole point: the limit-of-sums *definition* and the antiderivative *shortcut* are two routes to the same number. The Fundamental Theorem is not a different rule for area—it is the theorem that spares us this computation every time.

2 Antiderivatives and the Fundamental Theorem of Calculus

2.1 The antiderivative and the indefinite integral

Differentiation turns F into $f = F'$. Integration, in its second guise, asks the reverse question: given f, find a function whose derivative is f.

Definition 2.1 (Antiderivative and indefinite integral). An *antiderivative* of f is a function F with $F'(x) = f(x)$. Since the derivative of a constant is zero, if F is one antiderivative on an interval then so is $F + C$ for every constant C, and on that interval these are *all* of them. The whole family is the *indefinite integral*, written

$$\int f(x)\,\mathrm{d}x = F(x) + C,$$

where C is the *constant of integration*.

The indefinite integral carries no limits and is a *function* (strictly, a family of functions), in contrast to the definite integral, which is a number. Reversing the power rule $\frac{\mathrm{d}}{\mathrm{d}x}x^n = nx^{n-1}$ of Unit 7 gives the single most used formula of the subject.

Example 2.1 ***(The power rule, reversed).*** For any constant $n \neq -1$,

$$\int x^n\,\mathrm{d}x = \frac{x^{n+1}}{n+1} + C,$$

because differentiating the right-hand side returns x^n. Thus $\int x^4\,\mathrm{d}x = \frac{1}{5}x^5 + C$ and, writing roots and reciprocals as powers, $\int \sqrt{x}\,\mathrm{d}x = \int x^{1/2}\,\mathrm{d}x = \frac{2}{3}x^{3/2} + C$. The excluded case $n = -1$ is filled by $\int \frac{1}{x}\,\mathrm{d}x = \ln|x| + C$, the reverse of $\frac{\mathrm{d}}{\mathrm{d}x}\ln|x| = \frac{1}{x}$ (valid for $x \neq 0$; the absolute value is what makes it hold for negative x too).

2.2 The Fundamental Theorem of Calculus

The two guises of integration—signed area and antiderivative—look unrelated. The Fundamental Theorem of Calculus (FTC) says they are the same thing, and it comes in two parts. Throughout, f is continuous on $[a, b]$.

Theorem 2.1 (Fundamental Theorem of Calculus, Part 1). Define the "area-so-far" function $F(x) = \int_a^x f(t)\,\mathrm{d}t$ for $a \leq x \leq b$. Then F is differentiable and

$$F'(x) = f(x):$$

differentiating an integral with respect to its upper limit gives back the integrand.

The idea is short: increasing the upper limit by a sliver $\mathrm{d}x$ adds a strip of area $f(x)\,\mathrm{d}x$ to F, so $\mathrm{d}F = f(x)\,\mathrm{d}x$ and $F' = f$. The sliver picture can be made into an honest argument using only the continuity of f and a squeeze. Form the difference quotient of F at x. By the interval-splitting property of §1.2 (writing $\int_p^q f$ for $\int_p^q f(t)\,\mathrm{d}t$ when the variable is clear), the numerator is just the area of the thin strip between x and $x + h$,

$$\frac{F(x+h) - F(x)}{h} = \frac{1}{h}\left(\int_a^{x+h} f - \int_a^x f\right) = \frac{1}{h}\int_x^{x+h} f(t)\,\mathrm{d}t.$$

On the short interval between x and $x + h$ the continuous function f attains a smallest value m_h and a largest value M_h, so the strip's area is trapped between two rectangles. Taking $h > 0$ for the display,

$$m_h\,h \;\leq\; \int_x^{x+h} f(t)\,\mathrm{d}t \;\leq\; M_h\,h, \qquad \text{hence} \qquad m_h \;\leq\; \frac{F(x+h) - F(x)}{h} \;\leq\; M_h.$$

For $h < 0$ the integral and the width both change sign, and the two reversals cancel: the difference quotient is trapped between m_h and M_h exactly as before. So the squeeze holds for either sign of h, which is what the two-sided limit needs. As $h \to 0$ the interval shrinks to the single point x, and *because f is continuous* both the smallest and the largest value on it close in on $f(x)$: $m_h \to f(x)$ and $M_h \to f(x)$. The difference quotient is squeezed between them, so it too tends to $f(x)$; that is exactly $F'(x) = f(x)$. (This is where continuity, the hypothesis carried since Unit 7, does its work: without it m_h and M_h need not meet, and the area function can fail to be differentiable.)

Part 1 says every continuous function *has* an antiderivative—namely its own area function. Part 2 turns this into a computational tool.

Theorem 2.2 (Fundamental Theorem of Calculus, Part 2). If F is *any* antiderivative of f on $[a, b]$, then

$$\int_a^b f(x)\,\mathrm{d}x = F(b) - F(a).$$

Proof. By Part 1 the area function $\mathcal{F}(x) = \int_a^x f(t)\,\mathrm{d}t$ is an antiderivative of f, and any other antiderivative differs from it by a constant: $F(x) = \mathcal{F}(x) + C$. Then

$$F(b) - F(a) = \big(\mathcal{F}(b) + C\big) - \big(\mathcal{F}(a) + C\big) = \mathcal{F}(b) - \mathcal{F}(a) = \int_a^b f - \int_a^a f = \int_a^b f(x)\,\mathrm{d}x,$$

since $\int_a^a f = 0$. The constant C cancels—which is why we may use *any* antiderivative and drop the $+C$ when evaluating a definite integral. ■

The bracket notation $\big[F(x)\big]_a^b := F(b) - F(a)$ records the evaluation.

Example 2.2 ***(An FTC evaluation).*** To find $\displaystyle\int_2^4 x^2\,\mathrm{d}x$, take the antiderivative $F(x) = \frac{1}{3}x^3$ (no $+C$ needed) and subtract its values at the limits:

$$\int_2^4 x^2\,\mathrm{d}x = \left[\frac{x^3}{3}\right]_2^4 = \frac{4^3}{3} - \frac{2^3}{3} = \frac{64-8}{3} = \frac{56}{3}.$$

No partition, no limit: the antiderivative delivers in one line an exact area that the Riemann sums of Section 1 could only bracket—as we check on their own example below.

Example 2.3 ***(Net change from a rate).*** The integral's physical reading is "total accumulated change." Take the particle of Unit 7 with displacement $s(t) = t^3 - 6t^2 + 9t$ and velocity $v(t) = s'(t) = 3t^2 - 12t + 9$. Its net displacement over the first three seconds is the integral of the velocity,

$$\int_0^3 v(t)\,\mathrm{d}t = \Big[t^3 - 6t^2 + 9t\Big]_0^3 = (27 - 54 + 27) - 0 = 0.$$

The particle ends exactly where it began—it travels out and returns, and the signed integral records the cancellation. In general $\int_a^b F'(t)\,\mathrm{d}t = F(b) - F(a)$: integrating a rate of change over an interval recovers the net change in the quantity itself.

Example 2.4 ***(Part 1 in action).*** Part 1 lets us differentiate an integral without evaluating it. For instance

$$\frac{\mathrm{d}}{\mathrm{d}x}\int_0^x \sin(t^2)\,\mathrm{d}t = \sin(x^2),$$

even though the integral itself has no elementary antiderivative. The upper limit is simply substituted into the integrand.

Often the upper limit is not x itself but a *function* of x. Part 1 then combines with the chain rule of Unit 7. Write F for the area function $F(u) = \int_a^u f(t)\,\mathrm{d}t$, so that $F' = f$ by Part 1. An integral whose upper limit is $g(x)$ is the composite $G(x) = \int_a^{g(x)} f(t)\,\mathrm{d}t = F\big(g(x)\big)$, and differentiating the composite gives

$$\frac{\mathrm{d}}{\mathrm{d}x}\int_a^{g(x)} f(t)\,\mathrm{d}t = F'\big(g(x)\big)\,g'(x) = f\big(g(x)\big)\,g'(x):$$

substitute the upper limit into the integrand, then multiply by the derivative of that limit.

Example 2.5 ***(A variable limit that is a function).*** With another integrand that has no elementary antiderivative,

$$\frac{\mathrm{d}}{\mathrm{d}x}\int_1^{x^3} \mathrm{e}^{-t^2}\,\mathrm{d}t = \mathrm{e}^{-(x^3)^2}\cdot\frac{\mathrm{d}}{\mathrm{d}x}\big(x^3\big) = 3x^2\,\mathrm{e}^{-x^6}.$$

The constant lower limit 1 plays no role—only the upper limit is fed into the integrand. Should the *lower* limit be the variable instead, reverse the limits first to bring it on top, which flips the sign: $\frac{\mathrm{d}}{\mathrm{d}x}\int_x^b f(t)\,\mathrm{d}t = -f(x)$.

2.3 Properties of the definite integral

Because a definite integral is built from a sum and evaluated through an antiderivative, it inherits the linearity of both. Let f, g be continuous and c a constant.

Theorem 2.3 (Properties of the definite integral).

$$\int_a^b (f\pm g) = \int_a^b f \pm \int_a^b g, \qquad \int_a^b c\,f = c\int_a^b f \qquad \text{(linearity)},$$

$$\int_a^a f = 0, \qquad \int_b^a f = -\int_a^b f \qquad \text{(reversing the limits flips the sign)},$$

$$\int_a^b f = \int_a^c f + \int_c^b f \qquad \text{(splitting the interval, for any } c\text{)}.$$

Proof. The last three were read off the Riemann sums in §1.2; Part 2 now confirms each of them in a line, with an antiderivative F. For the interval-splitting rule,

$$\int_a^c f + \int_c^b f = \big(F(c)-F(a)\big) + \big(F(b)-F(c)\big) = F(b)-F(a) = \int_a^b f,$$

the middle term $F(c)$ cancelling. The reversal rule is $F(a)-F(b) = -\big(F(b)-F(a)\big)$. For linearity, if $F' = f$ and $G' = g$ then $F\pm G$ and cF are antiderivatives of $f\pm g$ and cf (Unit 7), and Part 2 does the rest. ∎

Example 2.6 ***(Linearity at work).***

$$\int_0^1 \left(3x^2 - 4x + 5\right)\,\mathrm{d}x = \left[x^3 - 2x^2 + 5x\right]_0^1 = (1 - 2 + 5) - 0 = 4.$$

The integral of a sum is the sum of the integrals, and each constant multiple comes straight out. As a second instance, the area the rectangles of Section 1 were bracketing is now exact:

$$\int_0^3 \left(0.4 + 0.25x^2\right)\,\mathrm{d}x = \left[0.4\,x + \tfrac{1}{12}x^3\right]_0^3 = 1.2 + 2.25 = 3.45,$$

comfortably inside the bracket $[\,2.92,\ 4.04\,]$ found there.

Example 2.7 ***(Signed area versus total area).*** The Fundamental Theorem also settles the signed-area note of Section 1. The positive hump of $\sin x$ on $[0, \pi]$ has area

$$\int_0^{\pi} \sin x\,\mathrm{d}x = \left[-\cos x\right]_0^{\pi} = (-\cos\pi) - (-\cos 0) = 1 + 1 = 2,$$

while the dip on $[\pi, \frac{3\pi}{2}]$ contributes $\displaystyle\int_{\pi}^{3\pi/2} \sin x\,\mathrm{d}x = \left[-\cos x\right]_{\pi}^{3\pi/2} = 0 - 1 = -1$. Their sum is the net signed area $2 + (-1) = 1$ quoted there. The *total* area painted, however, is $2 + |-1| = 3$: to find a total area, integrate each piece between consecutive crossings of the axis and add the *magnitudes*.

3 Standard integrals and substitution

3.1 A table of standard integrals

Every entry below is one of the standard derivatives of Units 7–8 read from right to left; learning the derivative table therefore gives the integral table for free. Throughout, $a > 0$ is a constant and a single $+C$ is understood.

$f(x)$	$\int f(x)\,\mathrm{d}x$	$f(x)$	$\int f(x)\,\mathrm{d}x$
x^n $(n \neq -1)$	$\dfrac{x^{n+1}}{n+1}$	$\cos(ax)$	$\dfrac{1}{a}\sin(ax)$
$\dfrac{1}{x}$	$\ln|x|$	$\sec^2(ax)$	$\dfrac{1}{a}\tan(ax)$
e^{ax}	$\dfrac{1}{a}\mathrm{e}^{ax}$	$\dfrac{1}{a^2 + x^2}$	$\dfrac{1}{a}\arctan\dfrac{x}{a}$
$\sin(ax)$	$-\dfrac{1}{a}\cos(ax)$	$\dfrac{1}{\sqrt{a^2 - x^2}}$	$\arcsin\dfrac{x}{a}$

The last two come from the inverse-trigonometric derivatives of Unit 8 (with $a = 1$, $\arctan' x = \frac{1}{1+x^2}$ and $\arcsin' x = \frac{1}{\sqrt{1-x^2}}$) and are worth memorising in this direction, since they are the targets a substitution often aims for—for instance $\int \frac{\mathrm{d}x}{4+x^2} = \frac{1}{2}\arctan\frac{x}{2} + C$. With the table and linearity, any polynomial, exponential or simple trigonometric combination integrates on sight.

Example 3.1 ***(Straight from the table).***

$$\int \left(e^{2x} + \frac{3}{x} - \cos 5x\right) \mathrm{d}x = \tfrac{1}{2}e^{2x} + 3\ln|x| - \tfrac{1}{5}\sin 5x + C.$$

3.2 Integration by substitution

Most integrands are not in the table. Substitution is the technique that reshapes them until they are, and it is precisely the chain rule of Unit 7 run backwards. Recall the chain rule applied to a composite $F\big(g(x)\big)$:

$$\frac{\mathrm{d}}{\mathrm{d}x}F\big(g(x)\big) = F'\big(g(x)\big)\, g'(x).$$

Reading this as an antiderivative statement, with $F' = f$, gives the rule.

Theorem 3.1 (Integration by substitution). If $u = g(x)$ is differentiable and f is continuous, then

$$\int f\big(g(x)\big)\, g'(x)\ \mathrm{d}x = \int f(u)\ \mathrm{d}u.$$

In Leibniz's bookkeeping we set $u = g(x)$ and $\mathrm{d}u = g'(x)\ \mathrm{d}x$, substitute to turn the whole integral into one in u, integrate, and finally put $g(x)$ back in place of u. The art is choosing u to be the "inside" function whose derivative is also present (up to a constant).

Example 3.2 ***(A first substitution: the reverse chain rule).*** For $\int 2x\ \sin(x^2+3)\ \mathrm{d}x$, the inside function is $u = x^2 + 3$, whose derivative $2x$ sits ready in the integrand. With $\mathrm{d}u = 2x\ \mathrm{d}x$,

$$\int 2x\ \sin(x^2+3)\ \mathrm{d}x = \int \sin u\ \mathrm{d}u = -\cos u + C = -\cos(x^2+3) + C.$$

A quick differentiation of the answer, by the chain rule, recovers the integrand—always a good check.

Example 3.3 ***(Manufacturing the missing constant).*** For $\int x\, e^{x^2}\ \mathrm{d}x$, put $u = x^2$, so $\mathrm{d}u = 2x\ \mathrm{d}x$ and hence $x\ \mathrm{d}x = \frac{1}{2}\ \mathrm{d}u$. The integrand has only $x\ \mathrm{d}x$, not $2x\ \mathrm{d}x$, so we supply the factor and compensate:

$$\int x\, e^{x^2}\ \mathrm{d}x = \int e^u\, \tfrac{1}{2}\ \mathrm{d}u = \tfrac{1}{2}e^u + C = \tfrac{1}{2}e^{x^2} + C.$$

A constant can always be moved across the integral sign in this way; a function cannot.

Example 3.4 ***(When the substitution leaves an x behind).*** For $\int x\sqrt{x+1}\ \mathrm{d}x$ the natural choice is $u = x+1$, so $\mathrm{d}u = \mathrm{d}x$. The root becomes $\sqrt{u}$, but the stray factor x is not part of $\mathrm{d}u$—so we rewrite it in terms of u too, using $x = u - 1$:

$$\int x\sqrt{x+1}\ \mathrm{d}x = \int (u-1)\sqrt{u}\ \mathrm{d}u = \int \left(u^{3/2} - u^{1/2}\right)\ \mathrm{d}u = \tfrac{2}{5}u^{5/2} - \tfrac{2}{3}u^{3/2} + C.$$

Putting $u = x + 1$ back gives

$$\int x\sqrt{x+1}\,\mathrm{d}x = \tfrac{2}{5}(x+1)^{5/2} - \tfrac{2}{3}(x+1)^{3/2} + C.$$

The lesson: when a substitution does not absorb every x, solve $u = g(x)$ for x and eliminate the remainder—there is no need to abandon an otherwise good choice.

Example 3.5 ***(Where a table entry comes from: the missing*** $1/a$***).*** The table of §3.1 lists $\int \frac{\mathrm{d}x}{a^2+x^2} = \frac{1}{a}\arctan\frac{x}{a} + C$ as "worth memorising," but the factor $\frac{1}{a}$ looks like it appears from nowhere. Substitution shows exactly where it comes from. The inverse-trig derivative we need is $\arctan' u = \frac{1}{1+u^2}$, so we rescale the variable to turn $a^2 + x^2$ into $a^2(1+u^2)$: put $u = x/a$, giving $\mathrm{d}u = \mathrm{d}x/a$, i.e. $\mathrm{d}x = a\,\mathrm{d}u$. Then

$$\int \frac{\mathrm{d}x}{a^2+x^2} = \int \frac{a\,\mathrm{d}u}{a^2(1+u^2)} = \frac{1}{a}\int \frac{\mathrm{d}u}{1+u^2} = \frac{1}{a}\arctan u + C = \frac{1}{a}\arctan\frac{x}{a} + C.$$

The single power of a that survives—one from $\mathrm{d}x = a\,\mathrm{d}u$, two cancelled against a^2 in the denominator—is precisely the $\frac{1}{a}$ in the table. Differentiating back, $\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{a}\arctan\frac{x}{a}\right) = \frac{1}{a}\cdot\frac{1/a}{1+(x/a)^2} = \frac{1}{a^2+x^2}$, confirms it. The table rows are not magic to be memorised blindly: each is a substitution result like this one.

Note (The f'/f pattern gives a logarithm). Whenever the numerator is the derivative of the denominator, the substitution $u =$ denominator turns the integral into $\int \frac{\mathrm{d}u}{u} = \ln|u| + C$:

$$\int \frac{g'(x)}{g(x)}\,\mathrm{d}x = \ln\bigl|g(x)\bigr| + C.$$

For example, with $u = x^2 + 1$,

$$\int \frac{x}{x^2+1}\,\mathrm{d}x = \tfrac{1}{2}\ln(x^2+1) + C,$$

and with $u = \cos x$ (so $\mathrm{d}u = -\sin x\,\mathrm{d}x$, which supplies the leading minus),

$$\int \tan x\,\mathrm{d}x = \int \frac{\sin x}{\cos x}\,\mathrm{d}x = -\ln|\cos x| + C.$$

3.3 Substitution in a definite integral

For a definite integral there are two equally good routes. Either find the indefinite integral, return to x, and then apply the limits; or—usually cleaner—*change the limits* to match the new variable and never return to x at all. If $u = g(x)$ then as x runs from a to b, u runs from $g(a)$ to $g(b)$:

$$\int_{x=a}^{b} f\bigl(g(x)\bigr)\,g'(x)\,\mathrm{d}x = \int_{u=g(a)}^{g(b)} f(u)\,\mathrm{d}u.$$

Example 3.6 ***(The two routes, side by side).*** Evaluate $\displaystyle\int_0^2 x\,(x^2+1)^3\,\mathrm{d}x$ with $u = x^2 + 1$, so $\mathrm{d}u = 2x\,\mathrm{d}x$ and $x\,\mathrm{d}x = \frac{1}{2}\,\mathrm{d}u$.

Route 1 (indefinite first, then limits). Integrate in u, return to x, and only then apply the original x-limits:

$$\int x\,(x^2+1)^3\,\mathrm{d}x = \tfrac{1}{2}\int u^3\,\mathrm{d}u = \tfrac{1}{8}u^4 + C = \tfrac{1}{8}(x^2+1)^4 + C,$$

$$\int_0^2 x\,(x^2+1)^3\,\mathrm{d}x = \left[\tfrac{1}{8}(x^2+1)^4\right]_0^2 = \tfrac{1}{8}\left(5^4 - 1^4\right) = \tfrac{1}{8}(625-1) = 78.$$

Route 2 (change the limits). As $x : 0 \to 2$, $u = x^2+1$ runs $1 \to 5$, so we never return to x:

$$\int_0^2 x\,(x^2+1)^3\,\mathrm{d}x = \tfrac{1}{2}\int_1^5 u^3\,\mathrm{d}u = \tfrac{1}{2}\left[\tfrac{u^4}{4}\right]_1^5 = \tfrac{1}{8}(625-1) = 78.$$

Both give 78, as they must. Route 2 is usually the cleaner: converting the limits to u-values once removes the back-substitution step entirely.

Example 3.7 *(A substitution with changed limits).* Evaluate $\displaystyle\int_0^1 \frac{x}{1+x^2}\,\mathrm{d}x$. Take $u = 1+x^2$, so $\mathrm{d}u = 2x\,\mathrm{d}x$, i.e. $x\,\mathrm{d}x = \tfrac{1}{2}\,\mathrm{d}u$. The limits change with the variable:

$$x = 0 \;\Rightarrow\; u = 1, \qquad x = 1 \;\Rightarrow\; u = 2.$$

Hence

$$\int_0^1 \frac{x}{1+x^2}\,\mathrm{d}x = \tfrac{1}{2}\int_1^2 \frac{\mathrm{d}u}{u} = \tfrac{1}{2}\Big[\ln u\Big]_1^2 = \tfrac{1}{2}\left(\ln 2 - \ln 1\right) = \tfrac{1}{2}\ln 2.$$

Because the limits were converted to u-values, there was no need to write the answer back in terms of x before substituting them.

Example 3.8 *(A trigonometric changed-limit substitution).* For $\displaystyle\int_0^{\pi/2} \cos\alpha\,\sqrt{\sin\alpha}\,\mathrm{d}\alpha$, the inside function is $u = \sin\alpha$, with $\mathrm{d}u = \cos\alpha\,\mathrm{d}\alpha$; the limits become $u = \sin 0 = 0$ and $u = \sin\frac{\pi}{2} = 1$. The factor $\cos\alpha$ is exactly what $\mathrm{d}u$ absorbs:

$$\int_0^{\pi/2} \cos\alpha\,\sqrt{\sin\alpha}\,\mathrm{d}\alpha = \int_0^1 \sqrt{u}\,\mathrm{d}u = \left[\tfrac{2}{3}u^{3/2}\right]_0^1 = \tfrac{2}{3}.$$

Remark (Substitution is a search, not a recipe). There is no rule that names the right u in advance; you look for an inside function whose derivative is present up to a constant. If a trial substitution leaves behind a piece of the original variable that cannot be rewritten in terms of u (unlike the stray x above, which $x = u - 1$ rewrote), it was the wrong choice—abandon it and try another. Unit 10 adds the techniques for integrands where no single substitution suffices.

3.4 When there is no elementary antiderivative

Substitution—and the further techniques of Unit 10—will integrate a great many functions, but not all. It is a striking fact that some perfectly ordinary, continuous functions have *no antiderivative expressible in elementary terms*: no finite combination of powers, roots, exponentials, logarithms and trigonometric functions has them as its derivative. This is not a failure of cleverness but a theorem, due to Liouville.

Note (Some functions with no elementary antiderivative). Each integrand below is continuous, so by Part 1 of the Fundamental Theorem it *does* have an antiderivative—its own area function—yet that antiderivative cannot be written in closed form:

$$\int e^{-x^2}\,dx, \qquad \int \frac{\sin x}{x}\,dx, \qquad \int \sin(x^2)\,dx, \qquad \int \frac{e^x}{x}\,dx, \qquad \int \frac{1}{\ln x}\,dx.$$

The first is the bell curve behind the normal distribution; its area function is the *error function* erf up to a constant factor—the standard normalisation is $\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2}\,dt$—and it is central to statistics; it returns briefly in Unit 10. The rest name standard functions of physics and engineering—the sine integral Si (diffraction), the Fresnel integral (optics), and the exponential and logarithmic integrals.

When no elementary antiderivative is available the definite integral is still a perfectly good number—a signed area—and we compute it *numerically*, adding many thin strips exactly as in Section 1 but with a machine doing the arithmetic. The Riemann picture we started from is, in the end, also how the answer is obtained in practice.

Summary

Concept	**Key idea**
Definite integral	$\int_a^b f\ \mathrm{d}x = \lim_{\Delta x \to 0} \sum f(x_k)\Delta x$: a limit of Riemann sums; the net *signed* area between the graph and the x-axis. A number, with a dummy integration variable.
Antiderivative	F with $F' = f$; the *indefinite integral* $\int f\ \mathrm{d}x = F + C$ is the whole family (a function, not a number).
Power rule	$\int x^n\ \mathrm{d}x = \frac{x^{n+1}}{n+1} + C$ for $n \neq -1$; the case $n = -1$ is $\int \frac{1}{x}\ \mathrm{d}x = \ln\|x\| + C$.
FTC Part 1	$\frac{\mathrm{d}}{\mathrm{d}x}\int_a^x f(t)\ \mathrm{d}t = f(x)$: every continuous f has an antiderivative (its area function).
FTC Part 2	$\int_a^b f\ \mathrm{d}x = F(b) - F(a) = \left[F\right]_a^b$ for *any* antiderivative F; the $+C$ cancels.
Properties	Linearity; $\int_a^a = 0$; $\int_b^a = -\int_a^b$; $\int_a^b = \int_a^c + \int_c^b$.
Standard integrals	Each is a standard derivative reversed: $\mathrm{e}^{ax} \to \frac{1}{a}\mathrm{e}^{ax}$, $\sin ax \to -\frac{1}{a}\cos ax$, $\frac{1}{a^2+x^2} \to \frac{1}{a}\arctan\frac{x}{a}$, $\frac{1}{\sqrt{a^2-x^2}} \to \arcsin\frac{x}{a}$.
Substitution	The chain rule reversed: $u = g(x)$, $\mathrm{d}u = g'(x)\,\mathrm{d}x$ gives $\int f(g)g'\ \mathrm{d}x = \int f(u)\ \mathrm{d}u$. Move constants across the sign, not functions.
f'/f pattern	$\int \frac{g'}{g}\ \mathrm{d}x = \ln\|g\| + C$.
Definite substitution	Change the limits to u-values: $\int_{x=a}^{b} f(g(x))\,g'(x)\ \mathrm{d}x = \int_{u=g(a)}^{g(b)} f(u)\ \mathrm{d}u$; then no need to return to x.
No closed form	Some continuous functions (e^{-x^2}, $\frac{\sin x}{x}$, $\sin x^2$, …) have *no* elementary antiderivative (Liouville); the definite integral is still a number, evaluated numerically.

Next unit: the heavier techniques—integration by parts (the product rule reversed), partial fractions and trigonometric methods—and the applications of the definite integral: areas between curves, volumes of revolution, average values, and the expectation of a probability distribution, including integrals that reach to infinity.

Companion material: a separate *Exercise Sheet 9* (with solutions) develops the practice problems for this unit.